Python Lists Summary

1. List Indexing, and Slicing

Indexing:

```
nums = [10, 20, 30, 40]
print(nums[0])   # 10
print(nums[-1])  # 40
```

Slicing:

```
print(nums[1:3])  # [20, 30]
print(nums[:2])   # [10, 20]
print(nums[::2])  # [10, 30] (every second element)
print(nums[::-1]) # [40, 30, 20, 10] (reverse list)
```

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2. Common List Methods

1. `append()` : Add element to end

```
nums.append(50)
print(nums)  # [10, 20, 30, 40, 50]
```

2. `extend()` : Add multiple elements

```
nums.extend([60, 70])
print(nums)  # [10, 20, 30, 40, 50, 60, 70]
```

3. `insert()` : Insert at specific index

```
nums.insert(2, 25)
print(nums)  # [10, 20, 25, 30, 40, 50, 60, 70]
```

4. `remove()` : Remove first occurrence

```
nums.remove(30)
print(nums)  # [10, 20, 25, 40, 50, 60, 70]
```

5. `pop()` : Remove by index (default: last)

```
nums.pop()
print(nums)  # [10, 20, 25, 40, 50, 60]
```

6. `index()` : Find index of element

```
print(nums.index(40))  # 3
```

7. `count()` : Count occurrences

```
print(nums.count(20))  # 1
```

8. `sort()` : Sort list

```
nums.sort()  # Ascending
nums.sort(reverse=True)  # Descending
```

9. `reverse()` : Reverse order

```
nums.reverse()
print(nums)  # [60, 50, 40, 25, 20, 10]
```

10. `copy()` : Copy list

```
nums_copy = nums.copy()
```

11. `clear()` : Remove all elements

```
nums.clear()
```

3. List Usecases in Codeforces

1. Finding Maximum & Minimum

```
nums = list(map(int, input().split()))  # Read input as a l
ist of integers
print(max(nums))
print(min(nums))
```

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2. Summing Elements

```
nums = list(map(int, input().split()))  # Read input as a l
ist of integers
print(sum(nums))
```

---

3. Reversing a List

```
numbers = list(map(int, input().split()))
numbers.reverse()

for i in numbers:
    print(i, end=' ')  # Rerversing on the same lin
```

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4. Checking for Duplicates

```
nums = list(map(int, input().split()))  # Read input as a l
ist of integers
print(len(nums) != len(set(nums)))  # True (has duplicates)
```

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5. Sorting Elements

```
nums = list(map(int, input().split()))  # Read input as a list of integers
nums.sort() # ascending
nums.sort(reverse=True) # descending
print(nums)
```

6. Finding Most Common Element

```
from collections import Counter

arr = list(map(int, input().split()))
counter = Counter(arr)
most_common = counter.most_common(1)[0][0]  # Most common element
print(most_common)
```

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7. Finding First Unique Element

```
from collections import Counter

def first_unique(arr):
    count = Counter(arr)
    for num in arr:
        if count[num] == 1:
            return num
    return -1

arr = list(map(int, input().split()))
print(first_unique(arr))
```
